

Abstract Algebra

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1 Categories

Definition 1.1. A category $\mathbf{C}$ consists of:

- a class $\text{Obj}(\mathbf{C})$ of objects; and
- for every A, B of $\mathbf{C}$, a set $\text{Hom}_{\mathbf{C}}(A, B)$ of morphisms

$\text{Hom}_{\mathbf{C}}(A, B)$ must satisfy the following properties:

- For every A of $\mathbf{C}$, there exists $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$ called the identity on A .
- If $f \in \text{Hom}_{\mathbf{C}}(A, B)$ and $g \in \text{Hom}_{\mathbf{C}}(B, C)$, there exists a morphism $gf \in \text{Hom}_{\mathbf{C}}(A, C)$.
- If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$, and $h \in \text{Hom}_{\mathbf{C}}(C, D)$, then $(hg)f = h(gf)$.
- If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, then $f1_A = f$, and $1_Bf = f$.

A diagram of a category can be made with nodes as objects and arrows as morphisms; a commutative diagram is one in which any path with the same starting and ending object is the same morphism.

Example 1.1. It is easy to see that the category of sets, deemed $\mathbf{Set}$, satisfies the definition.

Example 1.2. Suppose S is a set and $\sim$ is a relation on S that is reflexive and transitive. Define the following:

- objects: elements of S ,

- morphisms: if $a, b \in S$, then let $\text{Hom}(a, b)$ consist of (a, b) if $a \sim b$ and empty otherwise.

Example 1.3. Consider $\mathbb{Z}$ with the relation $\leq$. Then this satisfies the conditions for being a category.

Note that every diagram in this category is commutative. This is not true in general: **Set** has diagrams which do not commute.

Morphisms

Definition 1.2. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is an isomorphism if it has a two-sided inverse under composition, aka if there exists $g \in \text{Hom}_{\mathbf{C}}(B, A)$ such that $gf = 1_A$ and $fg = 1_B$.

For this definition to make sense the inverse g has to be unique. This turns out to be true.

Definition 1.3. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is a monomorphism if for all objects Z of $\mathbf{C}$ and all morphisms $\alpha', \alpha'' \in \text{Hom}_{\mathbf{C}}(Z, A)$, $f \circ \alpha' = f \circ \alpha'' \Rightarrow \alpha' = \alpha''$.

Monomorphisms are the categorical generalization of injective functions.

Definition 1.4. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is an epimorphism if for all objects Z of $\mathbf{C}$ and all morphisms $\beta', \beta'' \in \text{Hom}_{\mathbf{C}}(B, Z)$, $\beta' \circ f = \beta'' \circ f \Rightarrow \beta' = \beta''$.

Epimorphisms are the categorical generalization of surjective functions.

Universal Properties

Definition 1.5. An object I of $\mathbf{C}$ is initial in $\mathbf{C}$ if for every $A \in \mathbf{C}$ there exists exactly one morphism $I \rightarrow A$. Similarly, an object $F \in \mathbf{C}$ is final if for every object $A \in \mathbf{C}$ there exists exactly one morphism $A \rightarrow F$. An object that is either initial or final is called a terminal object.

Example 1.4. In **Set**, the empty set $\emptyset$ is initial (with the 'empty graph') as the function from $\emptyset$ to any other set.

Set also has final sets, namely the sets with a single element $\{p\}$. The unique function from any set to this set is the constant function.

Thus terminal objects do not have to be unique. However, they are unique up to isomorphism. The proof is very simple and will not be provided here. Essentially, no terminal object is more “special” than any other terminal object (of the same kind).

Universal Properties. We are now in a spot to introduce universal properties. We say a certain construction satisfies a universal property when it may be viewed as a terminal object of a category. For example, in the category **Set**, saying that $\emptyset$ is universal with respect to the property of mapping to sets is synonymous with the fact that $\emptyset$ is initial in **Set**.

Example 1.5. Let $\sim$ be an equivalence relation on a set A . We say that the quotient $A/\sim$ is universal with respect to the property of mapping A to a set such that equivalent elements have the same image. What does this mean?

We are considering functions $\phi : A \rightarrow Z$ such that $a' \sim a'' \Rightarrow \phi(a') = \phi(a'')$. Then, based on the previous construction of different ϕ functions as objects and commutative diagrams as morphisms, we can consider whether this category has an initial element. It is relatively simple to show that the canonical projection, defined as the function from $A \rightarrow A/\sim$, is an initial element in this category. This, specifically, is what we mean by our statement about the universal property.

2 Groups

Definition 2.1. Let G be a nonempty set, and let $\cdot : G \times G \rightarrow G$ be a binary operation on G . We deem the multiplication of two elements by application of this function, i.e. $\cdot(g, h) = g \cdot h$. The two-tuple $(G, \cdot)$ is a group if the multiplication function satisfies the following:

- $\cdot$ is associative, that is, $(g \cdot h) \cdot k = g \cdot (h \cdot k)$ for all $g, h, k \in G$.
- there exists an identity element $e_G \in G$ such that $g \cdot e_G = e_G \cdot g$ for all $g \in G$.
- for every element $g \in G$ there exists an inverse element; that is, $g \cdot h = e_G = h \cdot g$ for some $h \in G$.

If, in addition to these properties, the multiplication function satisfies commutativity, then the group is called an Abelian group.

Proposition 2.1. If $h \in G$ is an identity element, then $h = e_G$.

Proof. $h = e_G h = e_G$. □

Thus we can see that the identity element is unique.

Proposition 2.2. For all $a, g, h \in G$, $ga = ha \Rightarrow g = h$ and $ag = ah \Rightarrow g = h$. In other words, cancellation holds in groups.

Proof. $ga = ha \Rightarrow (ga)a^{-1} = (ha)a^{-1} \Rightarrow g(aa^{-1}) = h(aa^{-1}) \Rightarrow ge_G = he_G \Rightarrow g = h$.

The second implication follows similarly. □

Definition 2.2. The order of a group G is $|G|$. An element g has finite order if $g^n = e$ for some positive integer n . Then the order of g , denoted $|g|$, is this integer n . If n does not exist then $|g| = \infty$. j

Definition 2.3. For two groups (G, m_G) and (H, m_H) , a group homomorphism $\varphi : (G, m_G) \rightarrow (H, m_H)$ is a function between sets such that the group operations are preserved, that is, $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$.

We are now in a place to define the category of groups:

Definition 2.4. Define the objects of the category $\mathbf{Grp}$ to be groups, and for any two groups G, H , we define $\text{Hom}_{\mathbf{Grp}}(G, H)$ to be the set of group homomorphisms from G to H .

We also have the following:

Definition 2.5. A group homomorphism between G and H is a group isomorphism if it is also a bijection.

Definition 2.6. Let G be a group, and let H be another group where $H \subseteq G$. Then H is a subgroup of G if the inclusion function $i : H \rightarrow G$ is a group homomorphism.

In other words, a group H defined on a subset of the elements of G is a group only if the operation defined respects the underlying G operation.

Proposition 2.3. A nonempty subset H of a group G is a subgroup iff

$$ab^{-1} \in H \quad \text{for all } a, b \in H \quad (1)$$

Definition 2.7. A subgroup N of a group G is called normal if $gN = Ng$ for all $g \in G$.

The importance of normal subgroups is that we may form quotient groups from them:

Definition 2.8. Suppose N is a normal subgroup of G . Then we may form a group deemed G/N defined as follows:

- elements of the group are gN .
- group multiplication is defined by $(gN) \cdot (hN) = (ghN)$

It should be relatively trivial to verify that this definition forms a group.

Free Groups

We define a group $F(A)$ for a set A called the free group. The construction is based on a suitable universality property. We consider a category F^A where the objects are (j, G) where G is a group and $j : A \rightarrow G$ is a set function. Now let morphisms between objects $(j_1, G_1) \rightarrow (j_2, G_2)$ to be commutative diagrams of set functions where $\varphi \circ j_1 = j_2$ where φ is required to be a group homomorphism. Our construction of $F(A)$ will be finding a suitable group such that $F(A)$ is initial in F^A .

Our construction is as follows: let A' be a set isomorphic to A such that for every $a \in A$ there is an associated element $a^{-1} \in A'$. A word on the set A is an ordered tuple

$$(a_1, a_2, \dots, a_n) \tag{2}$$

where each a_i is an element of either A or A' . The set of words on A is called $W(A)$. Note that $w = ()$ is a valid word.

For this construction to make sense we must be able to reduce a word by deleting copies of any aa^{-1} . Thus define a reduction operator $r : W(A) \rightarrow W(A)$ that reduces the first occurrence of aa^{-1} in the word. Now we have the following result:

Theorem 2.1. If $w \in W(A)$ has length n , then $r^{\lfloor \frac{n}{2} \rfloor}(w)$ is a reduced word.

Proof. This result is easy if we see that we cannot decrease the length of a word w by more than $n/2$ times. $\square$

Now define another operator $R : W(A) \rightarrow W(A)$ to be the operator $r^{\lfloor \frac{n}{2} \rfloor}(w)$.

We now finally define the free group $F(A)$. We define the elements of $F(A)$ to precisely be the set of reduced words as defined above. Then define an operation on this set to be the following:

$$w \cdot w' = R(ww') \quad (3)$$

It is clear that this is a group operation.

Now we have the final statement:

Proposition 2.4. The pair $(j, F(A))$, where $j : A \rightarrow F(A)$ is defined as sending the element $a \in A$ to the word (a) , is initial in F^A ; that is, it satisfies the universal property for free groups on A .

We will not prove this but it should be evident with all the notation defined above.

Presentations

Definition 2.9. A presentation of a group G is an explicit isomorphism

$$G \cong \frac{F(A)}{R} \quad (4)$$

where A is a set and R is a subgroup of 'relations'. A presentation of a group is denoted (A, R) . A group is finitely presented if it admits a presentation where both A and R are finite.

For example, S_3 admits a presentation as a quotient of $F(x, y)$.

Isomorphism Theorems

Definition 2.10. For two subsets $A, B \subseteq G$, let $AB = \{ab \mid a \in A, b \in B\}$.

Proposition 2.5. Let H, K be subgroups of G , and assume H is normal in G . Then

- HK is a subgroup of G , and H is normal in HK ;
- $H \cap K$ is normal in K , and

$$\frac{HK}{H} \cong \frac{K}{H \cap K} \quad (5)$$

Lagrange's Theorem

Definition 2.11. The index of H in G , denoted $[G : H]$, is the number of elements $|GH|$ of GH when this is finite, and ∞ otherwise.

Proposition 2.6. If G is a finite group and $H \subseteq G$ is a subgroup, then $|G| = [G : H] \cdot |H|$.

Proof. We know that G is the disjoint union of $|G/H|$ distinct cosets gH , thus $|gH| = |H|$. $\square$

Group Actions

Definition 2.12. A group action of a group G on an object A of a category $\mathbf{C}$ is a homomorphism

$$\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A). \quad (6)$$

In other words, every element $g \in G$ transforms the object A into itself. The most natural category for $\mathbf{C}$ is obviously **Set**, but there are many more, including smooth manifolds, topological spaces, etc.

Definition 2.13. An action of a group G on an object A of a category $\mathbf{C}$ is faithful if $\sigma : G \rightarrow \text{Aut}_{\mathbf{C}}(A)$ is injective.

Example 2.1. Every group G acts naturally on its underlying set G ; that is, it transforms every element of G to another element of G , isomorphically of course.

Definition 2.14. An action of a group G on a set A is transitive if for all $a, b \in A$ there exists $g \in G$ such that $b = ga$.

Definition 2.15. The orbit of an element $a \in A$ under a group action in a group G is the set

$$O_G(a) = \{ga | g \in G\} \quad (7)$$

Definition 2.16. The stabilizer subgroup of $a \in A$ consists of the elements $g \in G$ that fix a :

$$\text{Stab}_G(a) = \{g \in G | ga = a\} \quad (8)$$

The orbit of an action forms a partition of the set A , and it is easily seen that the action of G is now transitive; additionally, the different orbits of an action form a partition of A .

Proposition 2.7. Every transitive left-action of G on a set A is isomorphic to the left multiplication of G on G/H , where G/H are the left cosets of subgroups H of G , for $H =$ the stabilizer of any $a \in A$.

We now have the orbit-stabilizer theorem:

Proposition 2.8. If G is a finite group and O is an orbit on the set A , then

$$|O| = \frac{|G|}{|\text{Stab}_G(a)|} \tag{9}$$

for any $a \in O$.